

Secret Key Generation Based on Manipulated Channel Measurement Matching

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Abstract—The physical layer secret key generation exploiting wireless channel reciprocity has demonstrated its viability and effectiveness in various wireless scenarios, such as the Internet of Things (IoT) network, mobile communication network, and industrial control system. Most of the existing studies rely on the quantization technique to convert channel measurements into secret bits for confidential communications. However, non-simultaneous packet exchanges in time-division duplex systems and noise effects usually induce inconsistent quantization results and mismatched secret bits. Although recent research has spent significant effort mitigating such non-reciprocity, it is still far from practical error-free key generation. Unlike previous quantization-based approaches, we take a different viewpoint to match the randomly manipulated (i.e., permuted or edited) channel measurements between a pair of users by minimizing their discrepancy holistically. Specifically, two novel secret key generation algorithms based on bipartite graph matching (BM-SKG) and edited sequence alignment (SA-SKG) are developed. BM-SKG allows two users to generate the same secret key based on the permutation order of channel measurements, while SA-SKG aims to align the edited channel measurements between a pair of users for secret key agreement. In both algorithms, one user can preset the secret key and embed encrypted messages in the exchanged data packets, which reduces communication overheads in key generation. Extensive experimental results show that both BM-SKG and SA-SKG algorithms achieve error-free key agreement on channel measurements at a low cost under various scenarios.

Index Terms—Device-to-device communication, physical layer key generation, secure communications.

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I. INTRODUCTION

WITH the proliferation of wireless networks, ensuring the security of wireless channels between pairwise devices has become an urgent demand nowadays due to their broadcast nature. For instance, malicious attacks, such as eavesdropping, information theft, unauthorized access, and others, result in an annual loss of around 84 million euros [1]. To secure wireless communication, traditional encryption schemes [2], [3] mainly rely on generating secret keys based on pre-shared information to protect user messages from illegal access by malicious devices. However, these schemes are not preferred for device-to-device communications in wireless networks owing to the lack of key management infrastructures and the limited resources of mobile devices, especially for Internet of Things networks (e.g., sensors, wearable devices, etc.). In order to overcome the limitations of traditional encryption schemes, many researchers turn to the inherent physical properties of wireless channels to extract a shared secret key between a pair of wireless transceivers [4], [5], [6], [7], [8], [9], [10], [11], [12]. With information-theory security and low complexity advantages, physical layer secret key generation becomes a promising security mechanism in wireless scenarios for low-energy devices, such as Internet of Things (IoT) networks, mobile communication networks, and industrial control systems.

For most physical layer secret key generation methods, quantization is the most critical step in achieving the secret key agreement over a reciprocal wireless channel. However, the non-simultaneous channel probing in the time-division duplex system and hardware impairment plus noise effects [6], [7], [8] would harm the channel reciprocity, causing inconsistent quantization results and mismatched secret bits between two users. Considerable efforts have been made to mitigate such non-reciprocity in the wireless channel, however, with limited improvement. For example, Patwari et al. [6] and Zeng et al. [7] apply multi-level adaptive quantization and multi-antenna diversity, respectively, to achieve high key generation rates, but the mismatch rates are unsatisfactory in various scenarios. Li et al. [13] provide a theoretical study of the preprocessing schemes for generating high-agreement uncorrelated secret keys, while Margelis et al. [12] adopt the discrete cosine transform (DCT) on channel observations to improve the consistency of quantization results. We delve further and find that most of the existing work treats each channel measurement independently during the quantization process. However, the presence of abrupt

changes in channel fading and unpredictable ambient noise in practice would make it difficult to ensure consistent quantization results between a pair of users. Moreover, in slowly varying environments, such ambient noise would severely harm channel reciprocity, resulting in consecutive bits of 0's and 1's and more inconsistent bits in the quantization results between two users. Even with information reconciliation schemes [14], [15], [16], it is still challenging to completely correct these erroneous bits due to the limited error correction capability of these schemes. Hence, an efficient and robust solution is necessary to eliminate non-reciprocity and achieve error-free secret key agreement.

Towards this end, we take a different viewpoint on performing secret key generation by matching the manipulated channel measurements between a pair of users and develop two novel secret key generation algorithms based on bipartite graph matching (i.e., BM-SKG) and edited sequence alignment (i.e., SA-SKG). Specifically, suppose a pair of users, Alice and Bob, seek a shared key establishment based on their channel measurements. Alice randomly shuffles her channel measurements and informs Bob of the shuffling order, and Bob shuffles his channel measurements in the same way without revealing the actual channel measurement values. The shuffled channel measurements introduce higher entropy while still maintaining channel reciprocity. Next, the shuffled channel measurements of both users are segmented into multiple channel episodes, which are then processed using our proposed BM-SKG and SA-SKG algorithms for secret key generation. In particular, BM-SKG applies a random permutation to Alice's channel episodes and shares the permuted channel episodes with Bob. Then BM-SKG models these channel episodes as a bipartite graph and infers the episode permutation order by solving a weighted bipartite graph matching problem, which minimizes the discrepancy between the permuted episodes and Bob's episodes. In SA-SKG, Alice randomly edits her channel episodes (i.e., inserts and deletes some episodes) and then informs Bob of the channel measurements after editing. Next, Bob identifies the edited positions and corresponding operations in Alice's edited episodes by calculating the edit distance from his channel measurements. Since it is proven that the channel measurements are unknown to attackers at a reasonable distance from the users (i.e., over half a wavelength), the manipulated channel measurement sequence is also unknown to the attackers, and the inferred permutation order (for BM-SKG) and the editing pattern (for SA-SKG) could serve for the secret key generation. Furthermore, both algorithms allow Alice to preset the secret key before completing key generation. The manipulated channel measurements can then be transmitted to Bob with the encrypted message in the data packet, reducing communication overheads in key generation.

Realizing the proposed secret key generation methods faces several challenges. In BM-SKG, as the length of the involved channel measurement sequences increases, calculating the weighted bipartite graph matching incurs high overhead and bit mismatch. Thus, choosing an appropriate weight metric and an efficient matching algorithm is critical to ensuring robust and practical secret key generation. Furthermore, the success of SA-SKG is built on the accurate identification of the position and type of editing operations in the manipulated episodes.

Therefore, selecting a suitable distance metric and pattern matching algorithm is also challenging. In addition, an efficient information reconciliation scheme is also required to rectify the mismatches when the occasional mismatched episodes occur in the secret key generation. The main contributions of our work are as follows:

- We take a new approach to significantly achieving a low bit error rate and a high key generation rate by matching randomly manipulated episodes in channel measurements instead of relying on the quantization process used in traditional key generation methods. Moreover, our algorithm allows the secret key to be preset by one user, which can save communication costs.
- Compared to our previous work with high computational complexity [17], we propose an improved bipartite graph matching algorithm that accelerates the matching process through edge pruning and fulfills a highly accurate key agreement.
- Also, we present a practical key generation algorithm based on edited sequence alignment, SA-SKG, which aims to identify the editing pattern through dynamic programming to achieve accurate and efficient key agreement.
- We develop an efficient information reconciliation scheme and a channel reuse mechanism to ensure fast and error-free key agreement with high entropy. Extensive experiments based on a real dataset demonstrate that our proposed algorithms can achieve efficient secret key agreement with a low bit error rate for various scenarios.

II. RELATED WORK

The past two decades have witnessed the progress of secret key generation that leverages the reciprocity property of wireless channels. Among various existing studies, quantization plays a crucial role in achieving a secret key agreement between two users by exploiting various physical layer characteristics of the wireless channel. Received signal strength (RSS) measurements, readily available in existing wireless infrastructures, are widely utilized to generate secret bits. For instance, some RSS-based methods rely on the temporal and spatial variations of the radio channel [5], [6], [18], [19], antenna diversity [7], and a mean-value quantization scheme [11] to generate secret bits. Margelis et al. [12] and Kumar et al. [20] adopt the discrete cosine transform and discrete wavelet packet transform on channel observations, respectively, to reduce the mismatches caused by quantization. However, these methods have limited key generation and key agreement rates, especially in static and slowly varying environments. To improve channel fluctuation in such environments, Fang et al. [21], Aldaghri et al. [22], [23], and Li et al. [24] introduce techniques such as constructing a manipulated channel, induced randomness exchange, and channel obfuscation, respectively. Through channel manipulation and randomization, these schemes improve key generation and key agreement rates. However, implementing these schemes requires customized hardware devices, which may result in high energy costs and limited availability compared to commercial off-the-shelf devices, thereby hindering their widespread adoption for

key generations. Recently, Zhao et al. [19] introduce an adaptive quantization algorithm that can select the optimal quantization parameters, but it does not have a high enough key agreement rate in dynamic environments.

The phase information of the channel response [25], [26] has been considered an alternative characteristic for enabling secret key generation. However, the channel phase may not always be reciprocal since the hardware characteristics vary among different devices, resulting in a high disagreement rate. Moreover, secret bits can also be extracted from deep fades in channel gain caused by multipath [27], but this method is based on theoretical analysis and only provides simulation results.

Existing studies have provided a richer source of secret information by utilizing channel state information (CSI). For instance, Liu et al. [28] show the feasibility of using CSI for secret key extraction. A practical and fast secret key extraction scheme [8] is also proposed by developing a channel gain complement mechanism to mitigate the impact of non-reciprocity on quantization, but the key agreement rate is still limited in some scenarios. Moreover, Xi et al. [29] propose a validation-recombination mechanism that combines information from all sub-carriers to be resilient to key cracking attacks, and Zhang et al. [30] leverage orthogonal frequency-division multiplexing access to extend key generation to multiple users. However, these key generation methods depend on effective quantization and are susceptible to channel non-reciprocity caused by abrupt changes and ambient noise. Recently, Guo et al. [16] propose a lightweight key generation scheme that combines moving average filtering (MAF) with threshold-free quantization. Although threshold-free quantization reduces the mismatched bits, MAF has a limited effect on eliminating the interference of ambient noise, resulting in the necessity of complicated information reconciliation. Unlike the aforementioned work, our approach takes a different perspective by matching the randomly manipulated episodes of channel measurements between two users to enable low-cost, fast, and error-free secret key generation.

III. FEASIBILITY STUDY

A. Problem Formulation

Suppose that a probing signal, denoted as S , is transmitted wirelessly between Alice and Bob. The sequence of channel measurements received by Alice is defined as $V^A = H^{BA} \times S + \Pi^A$, where H^{BA} and Π^A are the channel response and ambient noise observed by Alice, respectively. Similarly, the sequence of channel measurements received by Bob is expressed as $V^B = H^{AB} \times S + \Pi^B$, where H^{AB} and Π^B are the channel response and ambient noise observed by Bob, respectively. Typically, the ambient noises Π^A and Π^B follow an i.i.d. zero-mean Gaussian distribution.

According to the principle of channel reciprocity, when Alice and Bob exchange probing signals within the channel coherence time, the channel responses H^{BA} and H^{AB} , as well as V^A and V^B , are highly correlated. Based on the above theories, traditional secret key generation methods allow Alice and Bob to extract the same secret bits by quantizing each individual channel measurement in V^A and V^B , respectively. However,

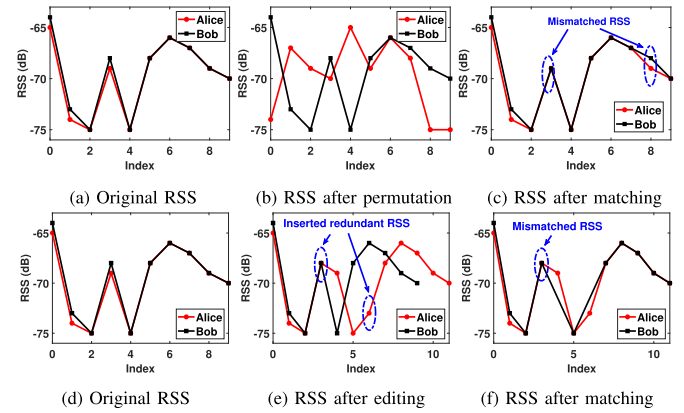


Fig. 1. Feasibility of using manipulated-based channel measurements matching to extract the secret bits.

the quantization with respect to each channel measurement independently renders V^A and V^B susceptible to non-simultaneous channel probing and random ambient noise [6] that are always present in wireless channels. The consequent inconsistent quantization outcomes and mismatched secret bits between two users degrade the performance of secret key generation in terms of accuracy and robustness.

To address the above issue, we propose to extract secret bits by finding a match between the manipulated channel measurements and the original channel measurements. Alice manipulates her channel measurements through random manipulation (i.e., permutation or editing) and then discloses these manipulated measurements to Bob. Despite the disruption of the original channel measurements, the reciprocal matching between Alice and Bob's channel measurements remains. If Bob can infer the manipulated pattern of Alice's channel measurements based on the disclosed channel measurements, the inferred pattern can serve as a shared secret between Alice and Bob for secret key generation. Moreover, even though the attackers are aware of some or all of the measurements, the random manipulation of channel measurements also diminishes their ability to correctly infer the order of the original measurements.

B. Preliminaries and Challenges

To verify the aforementioned speculation, we conduct preliminary experiments to examine the effectiveness of secret key generation through manipulated channel measurement matching. We deploy two laptops with an Intel 5300 NIC card operating in the 802.11n 2.4 GHz channel as a pair of legitimate users, Alice and Bob, who wish to establish a shared secret key. Then two RSS sequences, each consisting of 10 RSS measurements, are collected from the probe packets exchanged between Alice and Bob (shown in Fig. 1(a) and (d)). Next, the RSS sequence of Alice is permuted as shown in Fig. 1(b). At Bob's side, Bob conducts an exhaustive search for a minimum distance between the RSS sequences of Alice and Bob. Similarly, Alice can also manipulate the channel measurements by inserting some random measurements, as presented in Fig. 1(e). To infer the manipulation pattern, Bob enumerates all possible editing patterns based

on his channel measurements to match the manipulated RSS sequence from Alice. The matching results are shown in Fig. 1(c) and (f), where most of the RSS sequences find the correct match, except for very few RSS measurements as highlighted. Based on the above encouraging results, the remaining questions are how to efficiently infer the indices of the manipulated channel measurements and guarantee their uniqueness. To some extent, the inherent reciprocity and randomness of wireless channels would guarantee the uniqueness of the matching for the manipulated measurements. However, the proposed method will face some challenges in its practical implementation.

Robust Matching Algorithm. Ambient noise is the main cause of the mismatch among reciprocal channel measurements. Moreover, a steady channel measurement sequence with few sharp changes would magnify the impact of ambient noise, resulting in a severe degradation in key agreement performance. Furthermore, as shown in Fig. 1(c) and (f), it is unreliable to match individual channel measurements. Therefore, a robust matching algorithm with a sophisticated design should be developed to suppress the impact of ambient noise.

Efficient Searching Algorithm. Exhaustive searching with high computational complexity is not an optimal solution to infer the manipulation pattern. An efficient searching algorithm is required to achieve fast and practical matching among reciprocal channel measurements.

Effective Reconciliation Scheme. A robust matching algorithm can minimize the errors caused by ambient noise, but it cannot guarantee error-free key generation due to occasional non-reciprocity in the wireless channel. Therefore, it is essential to design an effective information reconciliation scheme for error-free key generation.

IV. ALGORITHM DESIGN

A. Basic Idea

The basic idea of our secret key generation approach is to generate an agreed-upon secret key between a pair of reciprocal users without involving error-prone quantization. In particular, the channel measurements from one user are manipulated using random operations. And based on the respective channel measurements, the user infers the manipulation pattern, which serves as the secret bits for key generation. During the manipulation, the channel measurements can also be arbitrarily specified by one user and be transmitted to the other user together with the encrypted message in the data packet.

Assuming a pair of users, Alice and Bob, intend to establish a secret key as shown in Figs. 2 and 3, they first collect their respective channel measurements, V^A and V^B , of length N , during *Channel Sampling* stage. Then Alice shuffles her channel measurements V^A with a random order $[k_1, k_2, \dots, k_N]$, which meets the entropy requirement (i.e., beyond a pre-defined threshold), and segments the shuffled measurements into M episodes as $\hat{P}^A = [p_1^A, p_2^A, \dots, p_M^A]$, where p_m^A is the m th episode with a length of $L = N/M$. Alice sends this random order to Bob, and Bob performs the same shuffling and segmentation to obtain M episodes $\hat{P}^B = [p_1^B, p_2^B, \dots, p_M^B]$. During this *Entropy-based Shuffling and Segmentation* stage, the shuffling process increases

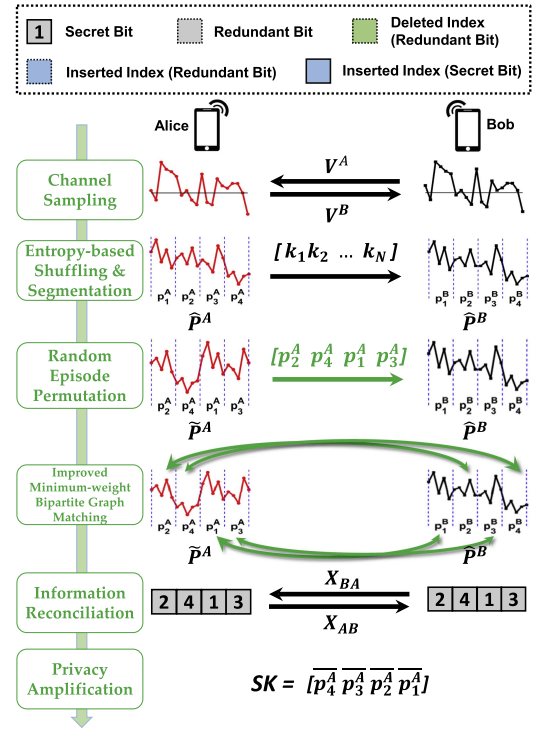


Fig. 2. BM-SKG algorithm flow.

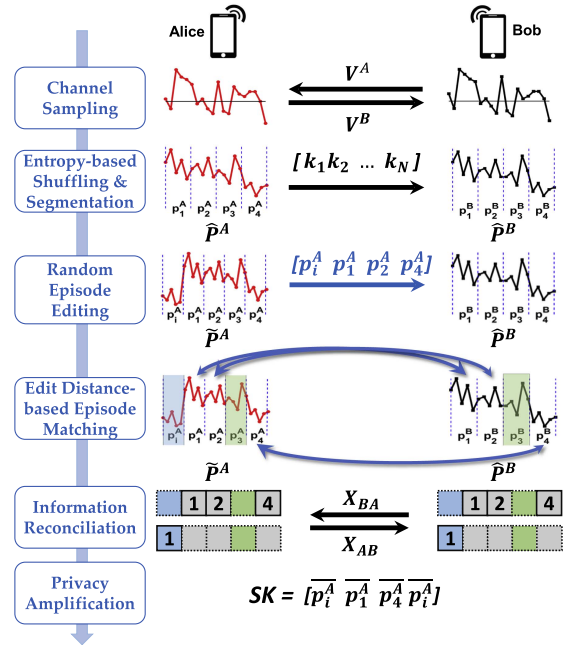


Fig. 3. SA-SKG algorithm flow.

the complexity (i.e., the variation) of the channel measurement sequence without destroying channel reciprocity, which helps reduce the ambiguity and mismatches caused by ambient noise. And the segmentation aims to increase the robustness of key agreement by involving more channel measurements for the following matching algorithms (BM-SKG and SA-SKG).

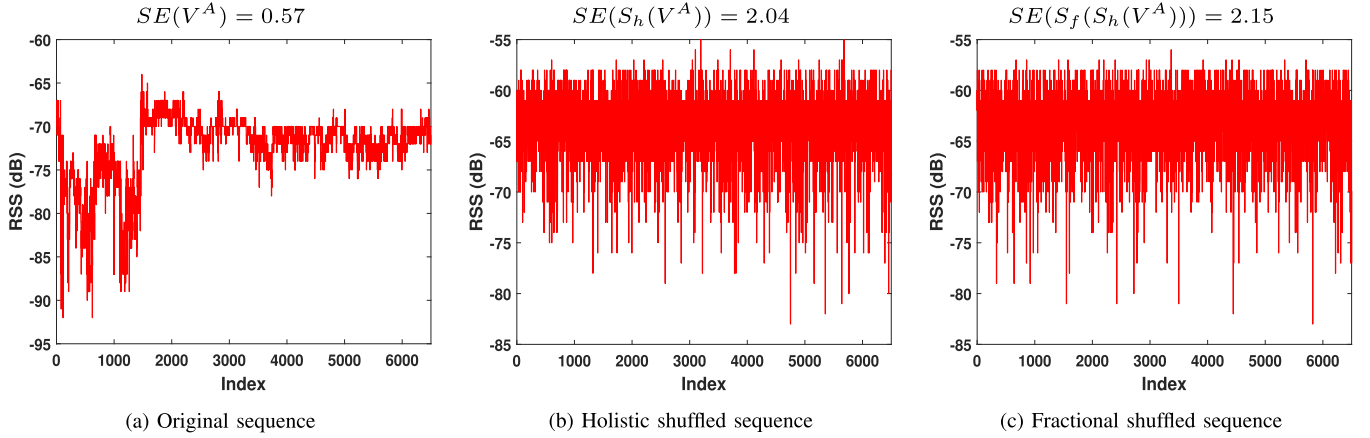


Fig. 4. Example of an RSS sequence after the entropy-based shuffling with evenly distributed channel interruption.

For BM-SKG, Alice applies *Random Episode Permutation* to the channel episodes $\hat{P}^A$ with a permutation order PO and constructs a new channel measurement sequence $\tilde{P}^A$. Then Alice sends $\tilde{P}^A$ to Bob via a public channel without revealing PO . By comparing $\tilde{P}^A$ with $\hat{P}^B$ through the *Improved Minimum-weight Bipartite Graph Matching* algorithm, Bob can infer the unique permutation order PO of $\tilde{P}^A$. Moreover, compared to our conference version [17], the improved algorithm enhances efficiency through edge pruning in the constructed bipartite graph during the matching process.

For SA-SKG, Alice manipulates her channel episodes $\hat{P}^A$ with *Random Episode Editing* (i.e., inserts and deletes some episodes) and creates a new channel measurement sequence $\tilde{P}^A$, which is then disclosed to Bob via a public channel. By performing the *Edit Distance-based Episode Matching* between $\tilde{P}^A$ and $\hat{P}^B$, Bob can infer the editing pattern in $\tilde{P}^A$, including the indices of unedited episodes I' and inserted episodes J' . Note that the actual indices of unedited episodes and inserted episodes are I and J , respectively.

In particular, for both algorithms, Alice can specify channel episodes $\tilde{P}^A$ with expected manipulation (i.e., permutation or editing) to preset the secret key. With the preset keys, Alice can embed encrypted messages in exchanged data packets during channel sampling, which can save communication costs in key generation. Finally, given the inferred manipulation pattern, Alice and Bob perform *Information Reconciliation and Privacy Amplification* to further eliminate occasional errors and generate a secret key with high randomness. The details of these components are elaborated as follows.

B. Bipartite Graph Matching-Based Key Generation (BM-SKG)

1) *Channel Sampling*: To initialize secret key generation, Alice and Bob exchange probe packets and extract corresponding channel measurements from the collected probe packets as $V^A = [v_1^A, v_2^A, \dots, v_N^A]$ and $V^B = [v_1^B, v_2^B, \dots, v_N^B]$. It is noted that each round of packet exchange needs to be conducted within the coherence time to ensure channel reciprocity. Once

a sufficient number of probe packets (e.g., RSS or CSI) are collected, the channel sampling process is complete.

2) *Entropy-Based Shuffling and Segmentation*: The theoretical basis of key generation is channel reciprocity between users, which is susceptible to non-simultaneous channel sampling in time-division duplex systems and ambient noise in practice. For instance, Fig. 4(a) shows channel measurement sequences involving steady episodes, where it is more difficult to produce superior reciprocity from them than from the relative dynamic episodes. Besides, ambient noise can further impair reciprocity among the steady episodes, adversely affecting the subsequent matching and key generation processes between Alice and Bob. Observe that the channel measurements of high complexity with abrupt changes can reduce ambiguity in the matching process that typically occurs in low-varying episodes. Hence, we perform an entropy-based shuffling to increase the complexity of the channel measurements and force the abrupt changes to be evenly distributed across the entire sequence of channel measurements.

Initially, Alice applies a random *holistic shuffling*, defined as S_h , to her channel measurement sequence $V^A = [v_1^A, v_2^A, \dots, v_N^A]$ and obtains $\hat{V}^A = S_h(V^A) = [\hat{v}_1^A, \hat{v}_2^A, \dots, \hat{v}_N^A]$. This shuffle only allows the abrupt changes to be randomly distributed among the sequence. To evaluate the complexity of the shuffled channel measurements, we adopt Sample Entropy (SE) [31], which provides insights into the complexity in the fluctuations of data sequences. A higher SE indicates that data sequences are less self-similar and more complex. For the detailed derivation of SE, refer to [31].

Given $SE(\hat{V}^A)$, Alice can repeatedly apply random shuffling to the channel measurement sequence $\hat{V}^A$ until its SE meets a predefined threshold. However, calculating the SE of the entire channel sequence takes a considerable amount of time. Therefore, we partition the channel sequence into multiple episodes and use *fractional shuffling*, defined as S_f , to evaluate the SE of each episode in channel measurements rather than that of the entire sequence. Specifically, both Alice and Bob segment their sequences $\hat{V}^A$ and V^B into multiple episodes $P^A = [\rho_1^A, \rho_2^A, \dots, \rho_M^A]$ and $P^B = [\rho_1^B, \rho_2^B, \dots, \rho_M^B]$ of a fixed length, where ρ_m^A or ρ_m^B denotes the m th episode in P^A or

P^B . Following this, Alice repeatedly applies random *fractional shuffling* (i.e., S_f) to each episode ρ_m^A in her channel measurement sequence P^A until $SE(\rho_m^A)$ of each segment is above a predefined threshold (i.e., typically twice the standard deviation of ρ_m^A). After fractional shuffling, the segmented sequence P^A involving steady episodes is transformed into a more irregular sequence $\hat{P}^A = [p_1^A, p_2^A, \dots, p_M^A]$. Fig. 4 shows that the SE of the channel measurement sequence rises from 0.57 to 2.15 after applying both holistic shuffling and fractional shuffling. Alice then combines two shuffling procedures (i.e., S_h and S_f) into one order $[k_1, k_2, \dots, k_N]$ and informs Bob of it without revealing her actual channel measurements, where $k_n \in [1, N]$ denotes the original index of channel measurement v_n^A in V^A . Upon receiving the order $[k_1, k_2, \dots, k_N]$, Bob shuffles V^B in the same order as Alice to V^A and obtains a new sequence $\hat{P}^B = [p_1^B, p_2^B, \dots, p_M^B]$. Due to channel reciprocity, the SE of $\hat{P}^B$ also meets the requirement.

Note that in the segmentation process, the length of the segmented episode should be carefully chosen to balance the key generation rate and bit error rate. A longer episode length ensures more robust matching results, which can lower the bit error rate. However, it also requires more channel measurements to represent a single secret bit, reducing the key generation rate. With a constant bit error rate, as we will see in Section VI, the key generation rate will decrease as the episode length increases. As a result, taking into account both the key generation rate and the bit error rate, we will choose a default episode length of 6 in this work.

3) *Random Episode Permutation*: After applying *Entropy-based Shuffling and Segmentation* to obtain the sequence $\hat{P}^A$, Alice performs *Random Episode Permutation* to facilitate the subsequent matching process. Specifically, she applies a random permutation to sequence $\hat{P}^A$ and creates a new channel measurement sequence $\tilde{P}^A = [p_{\kappa_1}^A, p_{\kappa_2}^A, \dots, p_{\kappa_M}^A]$, where $\kappa_m \in [1, M]$ is the original index of the episode $p_{\kappa_m}^A$ in $\hat{P}^A$. Then $\tilde{P}^A$ is broadcast via a public channel, where both Bob and potential attackers can hear it. Since the original channel measurement sequence was not made public, the episode permutation order $PO = [\kappa_1, \kappa_2, \dots, \kappa_M]$ remains unknown to the attackers and can serve as a shared secret between Alice and Bob.

Regarding the public sequence $\tilde{P}^A$, one non-negligible concern is whether an attacker can infer the permutation order PO by exploiting the correlation among its episodes. Since the entropy-based shuffling process increases the irregularity of the original channel measurement sequence V^A and the entropy of each segmented p_m^A , the correlation between any pair of episodes in $\tilde{P}^A$ is significantly reduced, leaving less chance for the attacker to infer the permutation order.

4) *Minimum-Weight Bipartite Graph Matching*: According to the channel reciprocity between Alice and Bob, each permuted episode $p_{\kappa_m}^A$ in the published $\tilde{P}^A$ should have a reciprocal episode p_{κ}^B in $\hat{P}^B$ with a similar pattern. As we propose to use the permutation order PO as the secret key, achieving key agreement between Alice and Bob is equivalent to determining the match between the episodes in $\tilde{P}^A$ and $\hat{P}^B$ with the minimum discrepancy.

Let $G = (V_A \cup V_B, E)$ be a bipartite graph with two disjoint vertex sets, V_A and V_B , and an edge set E . Each episode in $\tilde{P}^A$ or $\hat{P}^B$ serves as a vertex in V_A or V_B , respectively. An edge incident with a vertex κ_m in V_A and another one κ in V_B is written (κ_m, κ) (i.e., no edge connects any two vertices within $\tilde{P}^A$ or $\hat{P}^B$). Let $w_{A,B} : E \rightarrow \mathbb{R}_+$ be a weight function of the edges. Specifically, we choose the absolute value as the weight function of edges (κ_m, κ) , $w_{A,B}(\kappa_m, \kappa) = |p_{\kappa_m}^A - p_{\kappa}^B|$, where $\kappa_m, \kappa \in [1, M]$. We seek a minimum-weight bipartite graph matching in G , which consists of a set of vertex-disjoint edges in G . Due to channel reciprocity, there is always a minimum match in G that fulfills the reciprocal mapping between Alice's and Bob's channel measurement sequences. We formulate this problem as an integer linear programming (ILP) problem, aiming to minimize the sum of the associated edge weights:

$$\begin{aligned} \min \quad & \sum_{\kappa_m=\kappa_1}^{\kappa_M} \sum_{\kappa=1}^M w_{A,B}(\kappa_m, \kappa) \cdot x_{A,B}(\kappa_m, \kappa), \\ \text{s.t.} \quad & \sum_{\kappa_m=\kappa_1}^{\kappa_M} x_{A,B}(\kappa_m, \kappa) = 1, \kappa \in [1, M], \\ & \sum_{\kappa=1}^M x_{A,B}(\kappa_m, \kappa) = 1, \kappa_m \in [\kappa_1, \kappa_M], \\ & x_{A,B}(\kappa_m, \kappa) \in \{0, 1\}, \kappa_m \in [\kappa_1, \kappa_M], \kappa \in [1, M], \quad (1) \end{aligned}$$

where $x_{A,B}(\kappa_m, \kappa) = 1$ if $\kappa_m \in V_A$ is associated with $\kappa \in V_B$ and 0 otherwise. The optimal objective function value of the ILP problem must be at most the optimal value of the sum of weighted matches. To satisfy the complementary slackness condition $x_{A,B}(\kappa_m, \kappa) \in \{0, 1\}$, we construct a feasible solution to the dual of (1) with a value equal to the sum of weighted matching output by the algorithm as:

$$\begin{aligned} \max \quad & \sum_{a=\kappa_1}^{\kappa_M} y(a) + \sum_{b=1}^M y(b), \\ \text{s.t.} \quad & y(a) + y(b) \leq w_{A,B}(\kappa_m, \kappa), \\ & a \in [\kappa_1, \kappa_M], b \in [1, M]. \quad (2) \end{aligned}$$

It indicates that the minimum sum of matched edge weights of any feasible primal solution in (1) (i.e., a perfect matching Λ) is at least as great as the value of any feasible dual solution in (2). That is, for any dual feasible solution y , if we can find the minimum weight of the perfect matching Λ using only tight edges (i.e., an edge (a, b) is tight if $y(a) + y(b) = w_{A,B}(a, b)$), Λ must be optimal.

To solve the ILP problem, we maintain a feasible dual solution y and try to find a close-to-primal-feasible solution (i.e., matching Λ) that satisfies complementary slackness. We start with an arbitrary feasible dual solution y and find a maximum-cardinality matching Λ' that uses only tight edges. If Λ' is perfect, we are done; otherwise, we update our dual solution and repeat the process until an optimal solution is found. The overall complexity is bounded by $O(n(|E| + n \log n)) = O(n^3)$ [32] in

a complete bipartite graph, where $n = |\mathbf{V}_A| + |\mathbf{V}_B| = |\tilde{P}^A| + |\tilde{P}^B|$ is the number of vertices.

5) *Improved Minimum-Weight Bipartite Graph Matching*: The minimum-weight bipartite graph matching in a complete weighted bipartite graph has a complexity of $O(n^3)$. However, as more channel episodes are involved, inferring the permutation order will incur a high time overhead that may not be affordable for many mobile devices on wireless networks. To address this issue, we propose an improved algorithm to perform minimum-weight bipartite graph matching by pruning the edge set $\mathbf{E}$ in the graph $\mathbf{G}$. We first sort the edge weights $w_{A,B}(\kappa_m, \kappa)$ in increasing order for each vertex $p_{\kappa_m}^A \in \mathbf{V}_A$ and choose the first $C \log n$ vertices $p_{\kappa}^B \in \mathbf{V}_B$, where C is a constant that balances the running time of the algorithm and its probability of generating a minimum-weight bipartite graph matching. Similarly, we sort the edge weights $w_{A,B}(\kappa_m, \kappa)$ for each vertex $p_{\kappa}^B \in \mathbf{V}_B$ and preserve the first $C \log n$ vertices $p_{\kappa_m}^A \in \mathbf{V}_A$. Thus, by pruning the edge set $\mathbf{E}$, we transform a complete weighted bipartite graph $\mathbf{G} = (\mathbf{V}_A \cup \mathbf{V}_B, \mathbf{E})$ into a sparse weighted bipartite graph $\mathbf{G}' = (\mathbf{V}_A \cup \mathbf{V}_B, \mathbf{E}')$, where $|\mathbf{E}'|$ is bounded by $2Cn \log n$.

To determine if the pruned edge set $\mathbf{E}'$ in the sparse weighted bipartite graph $\mathbf{G}'$ contains a perfect matching, we refer to the theorem by Erdős and Rényi [33], which provides a probability estimate of the existence of a perfect match. Assuming that each edge occurs independently in $\mathbf{G}'$ with a probability of $\frac{2C \log n}{n}$, this theorem ensures the remaining edge set consisting of $2Cn \log n$ edges suffices to cover a perfect matching with high probability (i.e., $Pr = e^{-2e^{-(2C-1) \log n}}$). As such, the number of involved edges is reduced from n^2 to $2Cn \log n$, and the running time of the minimum-weight bipartite graph matching is accordingly reduced from $O(n^3)$ to $O(n(|\mathbf{E}'| + n \log n)) = O(n^2 \log n)$. Our experiment results in Section VI demonstrate the efficiency of the improved algorithm, which reduces the running time by 94.60% compared to the basic minimum-weight bipartite graph matching when the number of channel measurement episodes is fixed at 256. After this matching process, the permutation order inferred by Bob is denoted as $PO' = [\kappa'_1, \kappa'_2, \dots, \kappa'_M]$.

C. Edited Sequence Alignment-Based Key Generation (SA-SKG)

The SA-SKG algorithm uses the same *Channel Sampling*, *Entropy-based Shuffling and Segmentation* steps as the BM-SKG algorithm. Thus, we will not describe these steps again.

1) *Random Episode Editing*: Given the segmented sequences $\tilde{P}^A = [p_1^A, p_2^A, \dots, p_M^A]$, Alice conducts the manipulation through multiple edits at the episode level to create an M_p length channel measurement sequence $\hat{P}^A = [p_{\kappa_1}^A, p_{\kappa_2}^A, \dots, p_{\kappa_{M_p}}^A]$. Generally, there are four basic editing operations for data sequences: insertion, deletion, substitution, and transposition. To simplify the problem, we only use insertions and deletions¹ when editing $\tilde{P}^A$:

- *Insertion*: Insert an episode $p_{\alpha'}^A$ after position α and obtain $[p_1^A, p_2^A, \dots, p_{\alpha}^A, p_{\alpha'}^A, p_{\alpha+1}^A, \dots, p_{M_p}^A]$;
- *Deletion*: Delete an episode p_{β}^A at position β and obtain $[p_1^A, p_2^A, \dots, p_{\beta-1}^A, p_{\beta+1}^A, \dots, p_{M_p}^A]$;

Next, Alice selects L_i indices $[\alpha_1, \alpha_2, \dots, \alpha_{L_i}]$ for insertions and L_d indices $[\beta_1, \beta_2, \dots, \beta_{L_d}]$ for deletions at random and non-repetitively, with each index independently following a uniform distribution in the range of 0 to M . She then randomly selects to insert or delete an episode in $\tilde{P}^A$ based on the above indices, where the inserted episodes are randomly chosen from $\tilde{P}^A$ in another round of key generation and independently follow the identical distribution as the original episodes. After completing all L_i insertions and L_d deletions, Alice obtains a new channel measurement sequence $\tilde{P}^A$ and sends it to Bob.

2) *Edit Distance-Based Episode Matching*: SA-SKG extracts secret bits by aligning the editing patterns between the edited channel measurement sequence $\tilde{P}^A$ and the unedited $\hat{P}^B$. Specifically, Bob infers the indices in $\hat{P}^B$ that correspond to the unedited episodes in $\tilde{P}^A$ together with the indices corresponding to the inserted episodes in $\tilde{P}^A$. Due to the channel reciprocity, each unedited episode in $\tilde{P}^A$ can always find a reciprocal episode $p_{m'}^B$ in $\hat{P}^B$, otherwise for the inserted episodes.

To accurately infer the editing pattern, we formulate a minimum edit distance problem based on $\tilde{P}^A$ and $\hat{P}^B$. The objective is to search for a series of editing operations that convert $\tilde{P}^A = [p_{\kappa_1}^A, p_{\kappa_2}^A, \dots, p_{\kappa_{M_p}}^A]$ to $\hat{P}^B = [p_1^B, p_2^B, \dots, p_M^B]$, where the summation $D(\tilde{P}^A, \hat{P}^B)$ (i.e., the edit distance between $\tilde{P}^A$ and $\hat{P}^B$) of editing costs is minimized. As mentioned before, we only consider the two editing operations of insertion and deletion, and each operation is associated with a positive cost (e.g., $C_{ins} = 1$ and $C_{del} = 2$). We choose the absolute value $\Delta_{A,B}(\kappa_m, m') = \sum_{l=1}^L |p_{\kappa_m, l}^A - p_{m', l}^B|$ as the distance metric between each episode $p_{\kappa_m}^A$ in $\tilde{P}^A$ and $p_{m'}^B$ in $\hat{P}^B$, where $p_{\kappa_m}^A$ consists of L channel measurement as $[p_{\kappa_m, 1}^A, p_{\kappa_m, 2}^A, \dots, p_{\kappa_m, L}^A]$ and $p_{m'}^B$ consists of $[p_{m', 1}^B, p_{m', 2}^B, \dots, p_{m', L}^B]$. If $\Delta_{A,B}(\kappa_m, m')$ is less than a predefined threshold δ , the two episodes $p_{\kappa_m}^A$ and $p_{m'}^B$ are considered to be of the same pattern, and the episode $p_{m'}^B$ corresponds to the unedited episode $p_{\kappa_m}^A$ in $\tilde{P}^A$.

To efficiently compute the edit distance $D(\tilde{P}^A, \hat{P}^B)$, we define $D(\tilde{P}_i^A, \hat{P}_j^B)$ as the edit distance from $\tilde{P}_i^A = [p_{\kappa_1}^A, p_{\kappa_2}^A, \dots, p_{\kappa_i}^A]$ to $\hat{P}_j^B = [p_1^B, p_2^B, \dots, p_j^B]$, where $i \in [0, M_p]$, $j \in [0, M]$. Then we exploit dynamic programming to formalize the recurrence of $D(\tilde{P}_i^A, \hat{P}_j^B)$:

$$D(\tilde{P}_i^A, \hat{P}_j^B) = \min \left\{ \begin{array}{l} D(\tilde{P}_{i-1}^A, \hat{P}_{j-1}^B) \\ D(\tilde{P}_{i-1}^A, \hat{P}_j^B) + C_{del} \\ D(\tilde{P}_i^A, \hat{P}_{j-1}^B) + C_{ins} \end{array} \right\}, \quad (3)$$

where $i \in [1, M_p]$ or $j \in [1, M]$ and the boundary value is defined as $D(\tilde{P}_0^A, \hat{P}_0^B) = 0$. To compute the edit distance $D(\tilde{P}^A, \hat{P}^B)$, each $D(\tilde{P}_i^A, \hat{P}_j^B)$ needs to be evaluated for $i \in [1, M_p]$ and $j \in [1, M]$, which terminates at $D(\tilde{P}_{M_p}^A, \hat{P}_M^B) = D(\tilde{P}^A, \hat{P}^B)$. Then, the series of editing operations corresponding to $D(\tilde{P}^A, \hat{P}^B)$ can be inferred from each $D(A_i, B_j)$

¹ Substitution and transposition can be replaced by insertion and deletion.

Algorithm 1: Calculate Edit Distance and Editing Operations.

Input: two sequences $\tilde{P}^A$ and $\tilde{P}^B$ of channel measurement, threshold δ , distance function $\Delta_{A,B}(\cdot, \cdot)$;

Output: edit distance $D(\tilde{P}^A, \tilde{P}^B)$ and editing operations;

```

1: Initialization:
2:  $m \leftarrow$  number of episodes in  $\tilde{P}^A$ ;
3:  $n \leftarrow$  number of episodes in  $\tilde{P}^B$ ;
4:  $op \leftarrow$  editing operation matrix of  $(m+1) \times (n+1)$ ;
5:  $d \leftarrow$  edit distance matrix of  $(m+1) \times (n+1)$ ;
6:  $e, i, d \leftarrow$  short for no change, insertion, and deletion;
7:  $C_{ins} = 1, C_{del} = 2 \leftarrow$  costs of insertion and deletion;
8: for  $i = 1$  to  $m+1$  do
9:    $op[i][0] = op[i-1][0] + 'd'$ ; ▷ Deletion
10:   $d[i][0] = d[i-1][0] + C_{del}$ ;
11: end for
12: for  $j = 1$  to  $n+1$  do
13:   $op[0][j] = op[0][j-1] + 'i'$ ; ▷ Insertion
14:   $d[0][j] = d[0][j-1] + C_{ins}$ ;
15: end for
16: for  $i = 1$  to  $m+1$  do ▷ Iteration by row
17:   for  $j = 1$  to  $n+1$  do ▷ Iteration by column
18:    if  $\Delta_{A,B}(i, j) < \delta$  then
19:      $op[i][j] = op[i-1][j-1] + 'e'$ ; ▷ No change
20:      $d[i][j] = d[i-1][j-1]$ ;
21:    else if  $d[i-1][j] + C_{del} < d[i][j-1] + C_{ins}$  then
22:      $op[i][j] = op[i-1][j] + 'd'$ ; ▷ Deletion;
23:      $d[i][j] = d[i-1][j] + C_{del}$ ;
24:    else
25:      $op[i][j] = op[i][j-1] + 'i'$ ; ▷ Insertion;
26:      $d[i][j] = d[i][j-1] + C_{ins}$ ;
27:    end if
28:   end for
29: end for
30: return  $d[m][n], op[m][n]$ ;

```

by backtracking a minimizing path from $D(\tilde{P}_{M_p}^A, \tilde{P}_M^B)$ to $D(\tilde{P}_0^A, \tilde{P}_0^B)$. The details of edit distance calculation and editing operations inference are elaborated in Algorithm 1. The time complexity is bounded by $O(\min(M_p, M)D)$ [34], where D is the edit distance. Afterward, Bob infers the indices $I' = [i'_1, i'_2, \dots, i'_{M_r}]$ and $J' = [j'_1, j'_2, \dots, j'_{L_i}]$ in his original episodes $\tilde{P}^B$ that corresponds to the unedited episodes and the inserted episodes in $\tilde{P}^A$, respectively, where M_r denotes the number of the unedited episodes. Since Alice also has the consistent indices $I||J = [i_1, i_2, \dots, i_{M_r}; j_1, j_2, \dots, j_{L_i}]$, which serve as a unique and confidential sequence between Alice and Bob and can be utilized for the subsequent key establishment.

Remark: Both of our algorithms face the potential concern that two similar episode patterns occurring in the channel episodes may result in ambiguity in the matching process. To mitigate this issue, Alice examines the presence of two or more episodes with similar patterns in the channel episodes

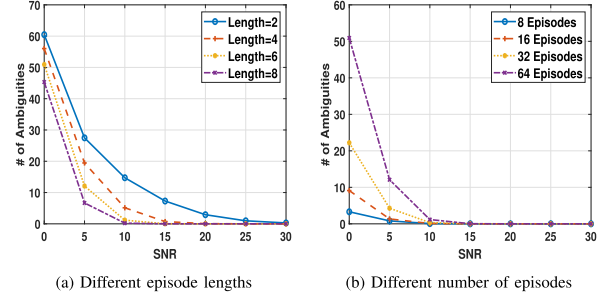


Fig. 5. Number of ambiguities with entropy-based shuffling and segmentation.

$\tilde{P}^A$ after the *Entropy-based Shuffling and Segmentation* stage. Specifically, Alice calculates the absolute differences between each pair of episodes in $\tilde{P}^A$ and repeatedly employs random *holistic shuffling* until the minimum absolute difference exceeds a predefined threshold. After this processing, each episode in $\tilde{P}^A$ acquires a certain degree of differentiation, ensuring that it can be effectively distinguished from one another in bipartite graph matching or edit distance-based episode matching. Therefore, the probability of encountering similar episodes and ambiguities is significantly diminished. To evaluate the probability of ambiguities arising during the matching process, we conduct a Monte Carlo simulation utilizing 100,000 pairs of measurements that follow a standard Gaussian distribution. We study the number of ambiguities under fixed parameters of 64 episodes and an episode length of 6. As depicted in Fig. 5, the probability of similar channel episode patterns leading to ambiguity diminishes with increasing episode length. Conversely, an increase in the number of episodes corresponds to an elevated probability of ambiguity. Moreover, it is notable that a higher signal-to-noise ratio (SNR) substantially reduces the occurrence of ambiguities. These strategies and observations collectively contribute to the reliability of both BM-SKG and SA-SKG algorithms against the potential challenge of ambiguity in the matching process.

D. Information Reconciliation and Privacy Amplification

The BM-SKG and SA-SKG algorithms can both achieve highly accurate key agreement, but information reconciliation is still essential for ensuring error-free key agreement. Instead of adding redundant bits to the secret key as in conventional methods [4], we propose to identify mismatched episodes with a two-round challenge-response message exchange between Alice and Bob as follows:

$$\text{Bob} \rightarrow \text{Alice} : X_{BA} = (\sigma')^{-1}(S^B \oplus R),$$

$$\text{Alice} \rightarrow \text{Bob} : X_{AB} = \sigma(X_{BA} \oplus S^A), \quad (4)$$

where $\oplus$ denotes the XOR operation. In BM-SKG, define the public sequence $\tilde{P}^A = [p_{\kappa_1}^A, \dots, p_{\kappa_M}^A] \triangleq [q_1, \dots, q_M]$, R is a random vector of length M , and $S^B = [\overline{q_{\kappa'_1}}, \dots, \overline{q_{\kappa'_M}}]$ and $S^A = [\overline{q_{\kappa_1}}, \dots, \overline{q_{\kappa_M}}]$, where $\overline{q_{\kappa'_m}}$ and $\overline{q_{\kappa_m}}$ are the respective average values of the κ'_m th and κ_m th episode in $\tilde{P}^A$. In SA-SKG, define the public sequence $\tilde{P}^A = [p_{\kappa_1}^A, \dots, p_{\kappa_{M_p}}^A] \triangleq [q_1, \dots, q_{M_p}]$, R is a random vector of

length $M_r + L_i$, $S^B = [\overline{q_{i_1}}, \dots, \overline{q_{i_{M_r}}}; \overline{q_{j_1}}, \dots, \overline{q_{j_{L_i}}}]$, and $S^A = [\overline{q_{i_1}}, \dots, \overline{q_{i_{M_r}}}; \overline{q_{j_1}}, \dots, \overline{q_{j_{L_i}}}]$, where each element in S^A and S^B are the respective average values of the corresponding episodes in $\hat{P}^A$. To facilitate the XOR operation, all elements in S^A and S^B and the random vector R are converted to binary numbers by rounding down and base conversion. For consistency of notation, let $\sigma(\cdot)$ be a permutation function that displaces a sequence with an order of $PO = [\kappa_1, \kappa_2, \dots, \kappa_M]$ in BM-SKG or an order of $I||J = [i_1, i_2, \dots, i_{M_r}; j_1, j_2, \dots, j_{L_i}]$ in SA-SKG. Besides, let $\sigma'(\cdot)$ be a permutation function that displaces a set with an order of $PO' = [\kappa'_1, \kappa'_2, \dots, \kappa'_M]$ or $I'||J' = [i'_1, i'_2, \dots, i'_{M_r}; j'_1, j'_2, \dots, j'_{L_i}]$. And $(\sigma')^{-1}(\cdot)$ represents the inverse permutation of $\sigma'(\cdot)$ with $\sigma'[(\sigma')^{-1}(S^B)] = S^B$. Since a permutation function of a finite set is a bijection from the set to itself, its inverse mapping (i.e., inverse permutation) always exists.

Next, Bob calculates $X_B = S^B \oplus R \oplus \sigma'(S^B)$. Ideally, Alice and Bob produce an identical permutation function such that $\sigma = \sigma'$ and $X_{AB} = X_B$. But there exist occasional mismatches between the original permutation σ and the actual one σ' due to non-simultaneous channel sampling and ambient noise. To determine the mismatched elements, Bob identifies the indices of unequal elements between X_{AB} and X_B as $\Phi = [\phi_1, \phi_2, \dots, \phi_S]$, where $\phi_s \in [1, M]$ in BM-SKG or $\phi_s \in [1, M_r + L_i]$ in SA-SKG. Considering a symmetric group of degree M , σ is a permutation of degree M and serves as an element in that group, and the indices Φ of unequal elements are also elements in that group. Since each permutation in the symmetric group can be decomposed into a composite mapping of several transpositions, and the number of mismatched indices is usually small, Bob can swap the mismatched elements in a few rounds until $X_{AB} = X_B$ holds. Eventually, the permutation order between Alice and Bob will be agreed upon (i.e., $\sigma = \sigma'$).

After information reconciliation, Alice and Bob will agree on an error-free secret key. But as a secret key, it also needs to satisfy a certain degree of randomness. Since $\hat{P}^A$ is published, Alice and Bob can enhance privacy and calculate the final secret key with the permutation operation as $SK = S^A = S^B$. Our experimental results, which will be presented in the experimental results section, show that SK achieves high randomness. Furthermore, to minimize information leakage, we utilize a universal hash function [35] for privacy amplification.

Security Analysis of Information Reconciliation: We only consider a passive attacker who conducts predictable channels or stalking attacks, which means he is only able to eavesdrop on reconciliation information without disrupting interactions. As we will analyze in Section V, the attacker is unaware of Alice's permutation order PO or Bob's permutation order PO' , indicating he has no knowledge of the permutation functions σ or σ' . Therefore, the attacker fails to calculate S^A by eavesdropping on X_{BA} and X_{AB} in (4). Besides, since the vector R in X_{BA} is randomly chosen by Bob, the attacker cannot obtain any additional information about S^B or the permutation function σ' . Therefore, our analysis demonstrates the security of information reconciliation against passive attackers.

E. Boosting Key Generation Rate With Channel Reuse

When the secret key is generated between two users in practice, they need to spend time performing *Channel Sampling*. Therefore, we propose a channel reuse mechanism to boost the efficiency of key generation. Our channel reuse mechanism can create multiple rounds of keys, and its security depends on the first round of key generation with our proposed BM-SKG and SA-SKG algorithms. Before each channel reuse, we manipulate the channel measurements to construct a new sequence and feed it into our BM-SKG or SA-SKG algorithm for another round of key generation. Specifically, given the secret key SK^i derived from the last round of key generation between Alice and Bob, we exploit SK^i as a seed value of the Fisher-Yates shuffle algorithm [36] to control the random shuffling of $\hat{P}^A$. Since SK^i is an agreed secret key between Alice and Bob, Bob can apply the same shuffling to $\hat{P}^B$ with the same seed as SK^i , without communicating with Alice. Moreover, since SK^i is unknown to the attackers, and there is no information exchange about the new shuffling order between Alice and Bob, the new shuffling order remains unknown to the attacker. Once $\hat{P}^A$ and $\hat{P}^B$ are shuffled again, the resulting sequences are used as input for the next round of BM-SKG or SA-SKG to generate SK^{i+1} .

Security Analysis of Channel Reuse Mechanism: Given a channel measurement sequence of length M , theoretically, we can use the same channel measurements $M!$ times with different permutation orders. However, many possible permutations have some measurements appearing in the same position, which may be exploited by attackers to infer the generated keys. In fact, a sequence of length M has at most M different permutations such that no element appears in the same position in any two of them. Typically, the first permutation that satisfies this condition is known as the derangement. The number of derangements of an M -element sequence can be calculated as $M! \sum_{i=0}^M \frac{(-1)^i}{i!} = \lfloor \frac{M!}{e} + \frac{1}{2} \rfloor$, where $M \geq 1$ and $\lfloor \cdot \rfloor$ is the floor function. Therefore, the probability that a randomly selected permutation is a derangement of the original sequence is $\sum_{i=0}^M \frac{(-1)^i}{i!} \approx \frac{1}{e}$, corresponding to a random selection only once. However, as the times of selected permutations increase, the probability that a randomly selected permutation will not have any elements in the same position as the selected permutation decreases dramatically. Therefore, for a channel measurement sequence consisting of M measurements, we specify that the sequence can be reused at most $\log_2(M)$ times to avoid potential security risks. For instance, 64 measurements are used to generate 6 different secret keys. Moreover, as we will see in Section VI, a small number of reuses will not compromise the randomness of the generated key, which indicates the feasibility of the channel reuse mechanism for improving the efficiency of key generation.

V. SECURITY ANALYSIS

A. Attack Model

In our attack model, we assume the attacker maintains an adequate distance from Alice and Bob (at least half of the

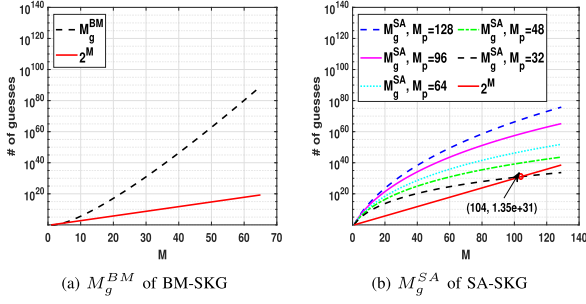


Fig. 6. Number of the attacker's guesses and the encryption strength of an M-bit key.

wavelength, e.g., 6.25 cm in the 2.4 GHz band) to avoid detection. Therefore, the attacker experiences largely uncorrelated fading channels, and its channel variations can thus be considered independent of legitimate users' channels [22], [24]. Moreover, it is assumed that the attacker has knowledge of the secret key generation algorithm between Alice and Bob. Specifically, we consider two types of attacks that have been identified as harmful to secret key generation in real-world scenarios.

Predictable Channel Attack [18]: A predictable channel attack can occur when Alice and Bob are stationary and the wireless channel between them is relatively stable. An attacker can cause predictable and desired changes in channel measurements between Alice and Bob by making planned movements. For example, if the line-of-sight (LOS) between Alice and Bob is intentionally blocked, the transmitted signal may experience sharp attenuation.

Stalking Attack [37]: A stalker can follow the trajectory of either Alice or Bob during the secret key establishment and eavesdrop on all legitimate communications exchanged between them. The stalker can measure the wireless channels between himself and Alice or Bob when they exchange probe packets.

We also assume that the attacker is not interested in disrupting the key generation. Specifically, we assume that the attacker does not modify the information exchanged between Alice and Bob or jam the communication channel between them. Moreover, we plan to address active attacks (e.g., jamming or man-in-the-middle attacks) in our future work.

B. Security Analysis of BM-SKG

Given a channel measurement sequence $\hat{P}^A$ consisting of M episodes in the BM-SKG algorithm, where each episode corresponds to an integer index within $[1, M]$, there are a total of $M!$ possible permutations for the generated secret key, which implies that the encryption strength will be $M!$. Fig. 6(a) shows that the number of attacker guesses $M_g^{BM} = M!$ varies with the length M of the episodes. For instance, $M = 64$ has an encryption strength much higher than a 256-bit key (i.e., $64! > 2^{256}$). If each episode of channel measurements is encoded as an ω -bit long binary number and each episode includes μ samples, the key generation rate is $\frac{\omega}{\mu}$. For instance, a 64-length episode of 6 samples is encoded as a $\log_2 64 = 6$ -bit long binary number, and the key generation rate will be $6/6 = 1$ bit/sample (bps).

During the key generation process, the information exchange between Alice and Bob does not directly or indirectly disclose any information related to the secret key (i.e., permutation order PO) to the attackers. Moreover, the randomness of the secret key is further strengthened by the privacy amplification process, which is verified through experimental studies. Therefore, the key agreement process based on our proposed BM-SKG algorithm is secure from external attackers. To crack the key, an attacker would have to randomly guess the secret bits, which is generally not feasible for most attackers.

When an attacker launches a predictable channel attack, the channel measurements published after random shuffling and permutation do not reveal any observable pattern, making it difficult for the attacker to obtain the correct permutation order. When a stalker attacker attempts to eavesdrop on the communication between Alice and Bob, he must stay at least half a wavelength away from Alice and Bob to avoid detection. Therefore, the attacker cannot infer the channel measurements or the secret key consistent with Alice's or Bob's.

C. Security Analysis of SA-SKG

In SA-SKG, Bob infers unedited episodes of length M in his original channel measurement sequence and inserted episodes of length M_p in the published sequence after random editing. For example, assume Bob's original sequence is $[p_1^B, p_2^B, p_3^B, p_4^B]$, and Alice's published sequence is $[p_i^A, p_1^A, p_2^A, p_4^A]$. This implies Alice inserts p_i^A into $[p_1^A, p_2^A, p_3^A, p_4^A]$ and deletes p_3^A . After performing *Edit Distance-based Episode Matching*, Bob infers the secret bits $I || J = [1, 2, 4; 1]$.

Therefore, to crack a key, an attacker needs to infer the indices I in the original sequence corresponding to the unedited episodes in the published episode and the indices J corresponding to the inserted episodes in the published episode. Since the location of edited episodes in the original sequence is generated randomly and non-repetitively, and the number of insertion or deletion operations (L_i or L_d) is also specified randomly, the length M_r of the unedited episodes cannot be accurately determined by the attacker. Specifically, the attacker first chooses an $M_r \in [0, M]$ and obtains $L_i = M_p - M_r$. Then, he needs to select the M_r -length unedited episodes from the original sequence consisting of M episodes and record their indices in the original sequence. Next, he chooses L_i -inserted episodes from the published sequence consisting of M_p episodes and records their indices in the published sequence. The number of guesses is denoted as

$$M_g^{SA} = \sum_{M_r=0}^M C_M^{M_r} \times C_{M_p}^{L_i}, M_p \in [M_r, 2M].$$

We now analyze the numeric relationship between M_g^{SA} and 2^M and present the following theorem.

Theorem 1: Let M be the length of episodes in the original sequence and M_p be the length of episodes in the published sequence. In SA-SKG, the number of attempts to guess the key exceeds 2^M if $M_p \geq M$.

Proof: When the length of episodes in the published sequence is greater than the length of episodes in the original sequence, we have $\sum_{M_r=0}^M C_M^{M_r} \times C_{M_p}^{L_i} = \sum_{M_r=0}^M C_M^{M_r} \times C_{M_p}^{M_r} \geq \sum_{M_r=0}^M C_M^{M_r} \times C_M^{M_r}$. Otherwise, M_p is less than M_r and there exists no combination $C_{M_p}^{M_r}$. According to the binomial theorem, we have:

$$\begin{aligned} \frac{\sum_{M_r=0}^M C_M^{M_r} \times C_M^{M_r}}{2^M} &= \frac{C_{2M}^M}{2^M} \\ &= \frac{2M}{2M} \cdot \frac{2M-1}{2(M-1)} \cdot \dots \cdot \frac{M+1}{2} > 1. \end{aligned} \quad (5)$$

Therefore, M_g^{SA} is greater than 2^M if $M_p \geq M$. $\square$

According to Theorem 1, SA-SKG only requires Alice to insert more episodes than deleted ones (i.e., $L_i > L_d$), with $M_p > M$, to ensure the generated key provides sufficient encryption strength.

Our numerical analysis verifies the validity of Theorem 1, as shown in Fig. 6(b) with a logarithmic scale. For instance, when $M = 64$ and $M_p = 96$, SA-SKG has an encryption strength greater than that of a 128-bit key. By varying the length M_p of episodes in the published sequence, we observe that M_g^{SA} increases with the growth of M_p . This is because as M_p increases, the second product component of M_g^{SA} also increases. Moreover, M_p should not be too small, such as 32, as indicated by the arrow, where M_g^{SA} is less than 2^M when M is greater than 104. Besides, we find that Theorem 1 provides a strict lower bound for M_p . In practice, Fig. 6 indicates that M_p should be no less than 48 to ensure an encryption strength greater than that of a 128-bit key (i.e., $M = 128$).

When an attacker launches a predictable channel attack, the random shuffling and editing processes make it challenging to infer the correct edited indices from the published sequence. Besides, the attacker must maintain a certain distance from Alice or Bob to launch an eavesdropping attack, thereby failing to extract the correct secret key.

D. Comparison of the Encryption Strength for BM-SKG and SA-SKG Algorithms

We conduct a comparison of the encryption strengths between SA-SKG and BM-SKG algorithms. In the key generation process, given M episodes in the channel measurement sequence, the attacker must guess the correct key from $M!$ possible permutations in BM-SKG. In contrast, the number of guesses in SA-SKG M_g^{SA} is bounded by C_{3M}^M . Thus, BM-SKG provides stronger encryption strength than SA-SKG does when $M \geq 14$.

While BM-SKG provides strong security, its weighted bipartite graph matching algorithm has a high time complexity, which increases the time cost of secret key generation. As noted in Section IV, the complexity of its optimal solution to provide improved minimum-weight bipartite graph matching is bounded by $O(M^2 \log M)$. On the other hand, SA-SKG has a time complexity of $O(\min(M_p, M)D)$ for calculating the edit distance and edited indices (D is the edit distance and $M_p \in [M_r, 2M]$), which provides a reasonable balance between security and efficiency. Therefore, BM-SKG is more appropriate

for applications with abundant computing resources and high security demands, while SA-SKG is suitable for time-sensitive applications.

VI. PERFORMANCE EVALUATION

A. Experimental Setup

To evaluate our proposed algorithms, we conducted experiments using a publicly available dataset created by G. Margelis et al. for their RSS-based key generation research [12]. This dataset was collected using the Internet of Things (IoT) devices and covers indoor and outdoor environments, with devices placed in both stationary and mobile positions. Specifically, the indoor environment (an office space) used the CC2650 radio [38] operating at 2.4 GHz, while the outdoor environment (near a busy university road) adopted the CC1310 radio [38] at 868 MHz. In both cases, three devices, Alice, Bob, and an attacker are employed, and each device collects at least 10,000 RSS measurement samples. The radios on these devices operated in half-duplex mode without affecting channel reciprocity.

B. Metrics

We present the experimental results using the following metrics, and all results are the average value based on the used RSS sequences.

Bit Error Rate (BER): It is the number of mismatched bits over the total number of secret bits generated. Note that the bit error rate is measured before information reconciliation.

Key Error Rate (KER): It is the number of mismatched keys over the number of generated secret keys across multiple rounds of key generation, measured before information reconciliation.

Key Generation Rate (KGR): It is defined as the number of secure bits per channel measurement for key agreement.

Randomness: It is used to assess the quality of keys, and we measure it using the NIST test [39].

C. Performance Evaluation

Secret Key Generation Performance: We first evaluate the performance of BM-SKG and SA-SKG algorithms by comparing them with six state-of-the-art quantization-based key generation algorithms, including the works of Zhao et al. [19], Fang et al. [21], Li et al. [24], Aldaghri et al. [22], [23], Guo et al. [16], and Margelis et al. [12], under different user motion statuses (i.e., mobile and stationary) and different environments (i.e., indoor and outdoor). Various secret key generation algorithms utilize one or more design parameters. For our experimental results, we use the following design parameters. In Zhao et al.'s algorithm, we set $p = 40$ and $\alpha = 0.1$ for a low BER. In Fang et al.'s algorithm, L is set to 5 for key retrieval. To strike a balance between BER and KGR, we opt for $\eta = 99.9\%$, a quantization level of 2, and $L_w = 64$ as suggested in [24]. In Aldaghri et al.'s algorithm, a quantization resolution of $\delta = 2$ is adopted to quantize channel measurements. Guo et al.'s algorithm is configured with $step = 4$ and $window = 8$ to achieve a trade-off between BER and KGR. In Margelis et al.'s algorithm, we employ the proposed SKYGlow.c algorithm to achieve a

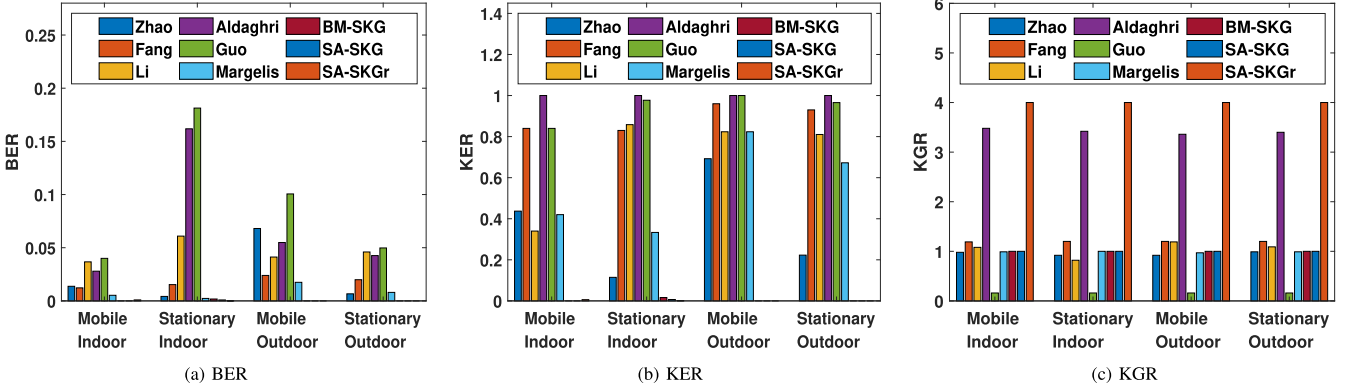


Fig. 7. Secret key generation performance comparison of different key generation algorithms.

lower BER. Notably, for the methods introduced in Fang et al.'s and Li et al.'s works, we only implement the quantization algorithm without involving channel manipulation or channel obfuscation.

As shown in Fig. 7, given the fixed number of episodes (i.e., 64) and episode length (i.e., 6), SA-SKG achieves a BER and KER of 0, except in indoor scenarios with stationary users, where the BER is 0.0009 and the KER is 0.0068. Meanwhile, BM-SKG, which only preserves the 16 largest edges for each vertex in a weighted bipartite graph, also achieves a BER and KER of 0, except in the same indoor scenario, where the BER is 0.0018 and the KER is 0.0162. However, Zhao et al.'s algorithm has relatively higher BER and KER, particularly in outdoor environments involving mobile users. Fang et al.'s, Li et al.'s, and Margelis et al.'s algorithms maintain relatively low BERs, while these outcomes are still much higher than ours and they fail to establish a consistent key between users. Guo et al.'s and Aldaghri et al.'s algorithms have comparatively high BERs and KERs in all scenarios, especially in environments with stationary users.

Furthermore, as indicated in Section V, 64 RSS episodes can achieve an encryption strength of over 128 bits in SA-SKG and over 256 bits in BM-SKG. If each episode index is encoded as a 6-bit binary number, both SA-SKG and BM-SKG can achieve a KGR of $6/6 = 1$ bps. In Fig. 7(c), we observe that the KGRs of Zhao et al.'s, Guo et al.'s, and Margelis et al.'s algorithms are lower than those of our algorithms. And the KGRs of the other algorithms (excluding Aldaghri et al.'s) modestly surpass ours. Overall, our SA-SKG and BM-SKG algorithms consistently outperform other quantization-based algorithms concerning the BERs and KERs, while maintaining comparable KGRs to most of these algorithms in all scenarios.

To evaluate the performance of the SA-SKG algorithm with channel reuse (short for SA-SKGr), we reuse the RSS sequence three times for a total of four rounds of key establishment. The results show that SA-SKGr has BERs and KERs close to 0 across all scenarios and outperforms other secret key generation algorithms in terms of KGR, reaching as high as 4 bps. For the BM-SKG algorithm with channel reuse (BM-SKGr), please refer to our conference version [17]. Since SA-SKGr and BM-SKGr have comparable BERs and KERs to SA-SKG and

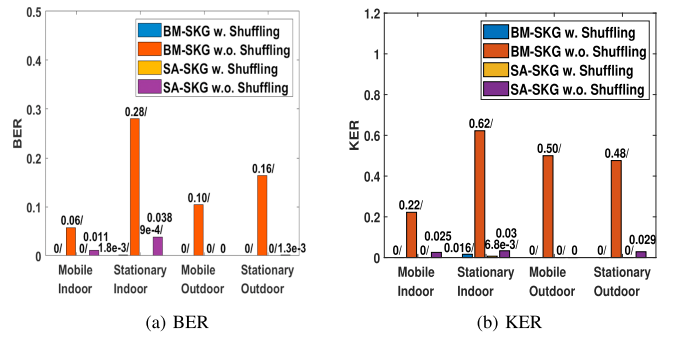


Fig. 8. Performance of BM-SKG and SA-SKG algorithms with and without the entropy-based shuffling.

BM-SKG, respectively, we only present the results for SA-SKG and BM-SKG in the following section.

Impact of Entropy-Based Shuffling: In this study, we investigate the impact of entropy-based shuffling on the performance of our BM-SKG and SA-SKG algorithms in terms of BER and KER. We consider an RSS episode length of 6, 64 RSS episodes, and 16 preserved edges for each vertex in a weighted bipartite graph. As mentioned in Section IV, entropy-based shuffling can increase the complexity of generated keys by disrupting the original RSS sequence order. Experimentally, we demonstrate that entropy-based shuffling highly improves the performance of our SA-SKG and BM-SKG algorithms. Our results show that BER is decreased from over 0.0013 to almost 0 in SA-SKG and from over 0.06 to almost 0 in BM-SKG for different motion statuses and environments, as shown in Fig. 8(a). Correspondingly, Fig. 8(b) shows that with the entropy-based shuffling, KER is reduced from over 0.025 to almost 0 in SA-SKG and from over 0.22 to nearly 0 in BM-SKG. We find that entropy-based shuffling provides more performance improvements to BM-SKG than SA-SKG because steady episodes without shuffling are more likely to produce ambiguity in a sparse bipartite graph matching. Overall, our results demonstrate the effectiveness of the entropy-based shuffling in BM-SKG and SA-SKG algorithms.

Impact of Episode Length: Next, we study how episode length affects the key generation performance of BM-SKG and

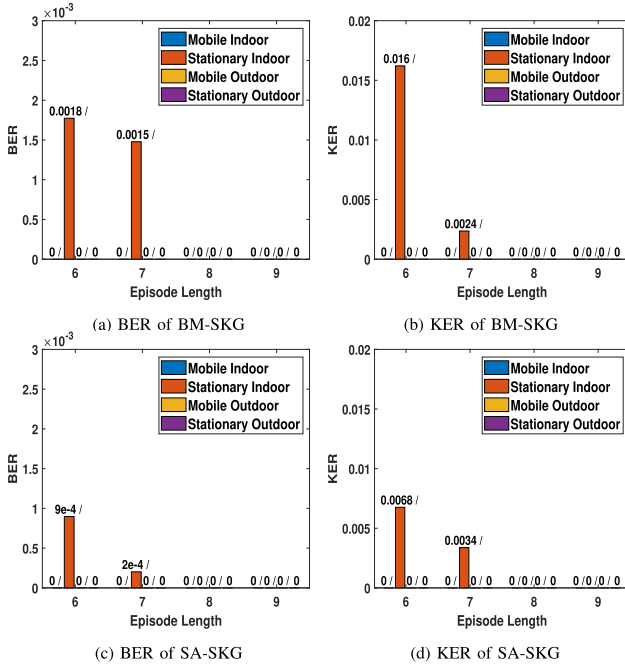


Fig. 9. Performance of BM-SKG and SA-SKG algorithms under the impact of different episode lengths.

SA-SKG. We set the number of RSS episodes used for key generation to 64 while varying the RSS episode length from 6 to 9. We use a fixed number of preserved edges (i.e., 16) for each vertex in BM-SKG and a fixed threshold δ in SA-SKG. When determining the number of preserved edges, episode length affects the weights used for bipartite graph matching. As the episode length increases, the probability of forming the same weights in a weighted bipartite graph decreases, resulting in performance improvements in the BERs and KERs. When setting the threshold δ , the length of the episode influences whether Alice and Bob can achieve the same edit distance. Short episodes increase the discrepancy between the original and edited episodes, resulting in unequal edit distances calculated by two users. In contrast, long episodes decrease the difference in RSS episodes, thereby lowering BERs and KERs. Fig. 9(a) and (c) show that for various environments, BER decreases from at most 0.0018 to 0 in BM-SKG and from 0.0009 to 0 in SA-SKG. Fig. 9(b) and (d) show that KER decreases from at most 0.016 to 0 in BM-SKG and from 0.0068 to 0 in SA-SKG for various environments. Therefore, it is critical to select an appropriate length (e.g., 8) of RSS episodes to ensure low BER and KER and to achieve error-free key agreement.

Impact of the Number of RSS Episodes: Using more RSS episodes can increase the probability of forming the same weights in the matching process of BM-SKG, and it increases the mismatch rate for key generation in SA-SKG due to the greater difference between the original and edited episodes. To examine this impact, we fix the episode length to 6 and calculate the BERs and KERs by varying the number of RSS episodes from 8 to 64, which is suitable for binary encoding. We also use the same threshold to calculate the edit distance in SA-SKG in different

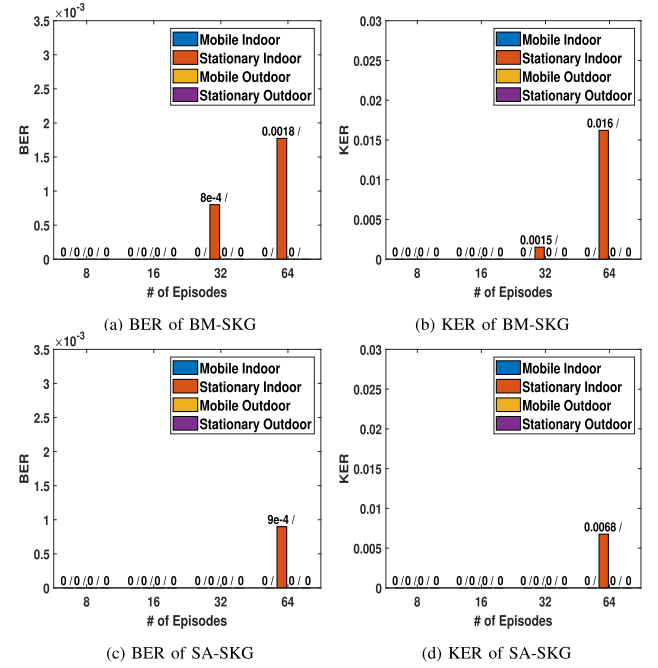


Fig. 10. Performance of BM-SKG and SA-SKG algorithms under the impact of different key lengths.

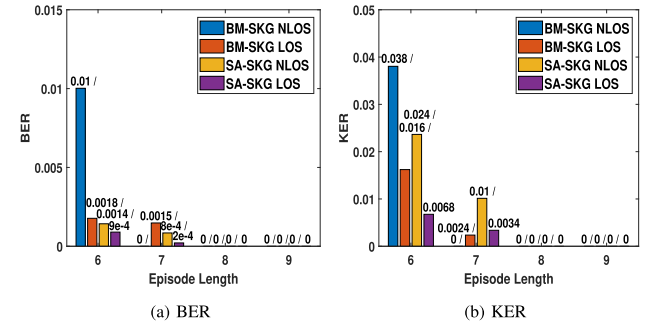


Fig. 11. Performance of BM-SKG and SA-SKG algorithms under the impact of LOS and NLOS channels.

environments. As shown in Fig. 10, the BERs and KERs for stationary and mobile users are close to 0 if the number of RSS episodes is less than 64 in both algorithms. Therefore, selecting a smaller number of RSS episodes is crucial to maintain a low mismatch rate and an acceptable KGR.

Impact of Line-of-Sight Channel: To further inspect the impact of the wireless propagation environment, we study the performance of line-of-sight (LOS) and non-line-of-sight (NLOS) channels using the dataset provided by [12]. The NLOS channel induces lower power levels at the receiver, leading to a lower SNR and making it susceptible to ambient noise, resulting in high BER and KER. Fig. 11 illustrates that as the episode length increases from 6 to 9, the BERs and KERs of both BM-SKG and SA-SKG algorithms for NLOS and LOS channels decrease to 0. These results demonstrate that the performance of secret key generation with LOS is superior to that with NLOS. We also find that increasing the episode length can suppress the channel's

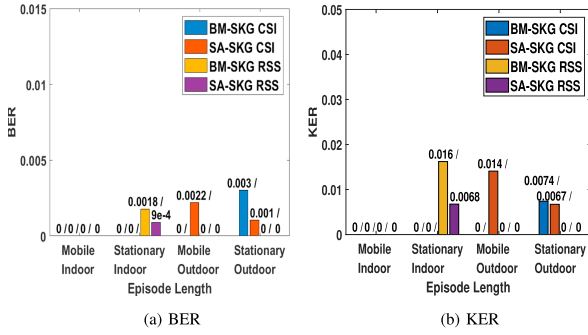


Fig. 12. Performance of BM-SKG and SA-SKG algorithms with CSI measurements.

impact. Additionally, the SA-SKG algorithm introduces more consistent changes between Alice and Bob through editing operations, enabling it to find a consistent editing pattern and, hence, perform better than the BM-SKG algorithm.

Extension to CSI Measurements: Our approach can easily be extended to use CSI measurements for secret key generation. To verify it, we conducted real-world experiments employing two USRP B210 devices placed in stationary and mobile positions, operating in the 802.11n 5 GHz channel. Due to the inter-subcarrier correlations inherent in CSI measurements [9], we pick 3 non-adjacent subcarriers to construct CSI sequences for key generation. We collect CSI measurements in indoor and outdoor environments. Fig. 12 shows the performance of our algorithms using CSI and RSS measurements to generate secret keys, where the episode numbers for both RSS and CSI measurements are kept constant at 64, and the episode length is fixed at 6. We find that the key generation using CSI measurements exhibits a low BER and KER, which is comparable to that of RSS measurements.

Time Cost: Notably, since our algorithms allow Alice to preset the secret key before completing key generation, the encrypted message can be carried in data packets exchanged during channel sampling and random episode permutation stages, lowering the time required for key generation. Consequently, we only analyze the time cost of our proposed algorithms without considering channel sampling. We vary the number of RSS episodes n from 16 to 256 for both algorithms, which corresponds to key lengths from 64 to 2048 bits while preserving $2 \log n$ edges for each vertex in the weighted bipartite graph matching of BM-SKG. Fig. 13(a) presents the time costs of SA-SKG, BM-SKG, and BM-SKG with a complete bipartite graph (BM-SKGc). It shows that the time cost of SA-SKG increases from 0.0003 sec to 0.1895 sec, and the time cost of BM-SKG with sparse bipartite increases from 0.0020 sec to 0.3717 sec for one key generation. Notably, at the cryptographic strength of 128 bits, the time cost for 64 episodes in BM-SKG is about 27 msec, and that for 64 episodes in SA-SKG is around 9 msec. Furthermore, Fig. 13(a) shows the time cost of BM-SKGc surges with the increase in the key length, whereas the time consumption of SA-SKG and BM-SKG is not significant, even for a key of 2048 length. These results demonstrate the effectiveness of our improved weighted bipartite graph matching algorithm and the

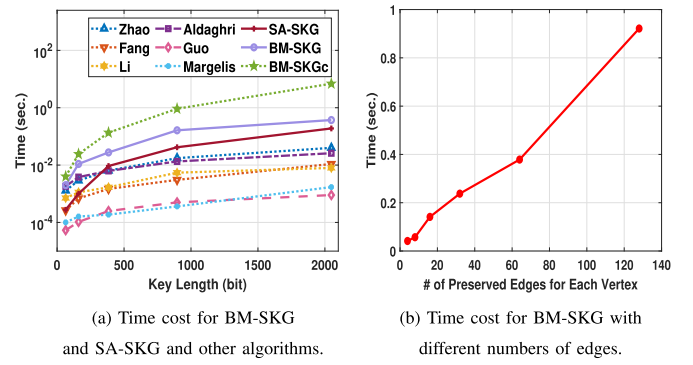


Fig. 13. Time cost for secret key generation.

analysis of time complexity for SA-SKG and BM-SKG. Besides, we compare the time cost of generating a secret key using the recent quantization-based algorithms with our algorithms. We observe that quantization-based algorithms exhibit shorter key generation times as they do not involve a relatively complex matching process. Nevertheless, it is noted that our algorithms, while incurring a little more time, deliver substantially improved accuracy in key generation, thereby demonstrating their practical applicability.

Fig. 13(b) shows that when the number of RSS episodes is fixed at 128, the time cost in BM-SKG increases with the number of preserved edges, indicating that the preserved edge number is the primary factor for the time cost in weighted bipartite graph matching. Therefore, it is possible to choose a small number of preserved edges (e.g., $2 \log n$) and perform an improved weighted bipartite graph matching in a sparse graph considering an acceptable time cost.

Randomness Test with NIST: To ensure that the generated secret key is sufficiently random, we employ the standard randomness test suite from NIST [39] to examine the randomness of the final key sequence generated by BM-SKG and SA-SKG after privacy amplification. We perform nine NIST tests and calculate the probability value, known as the p-value, for each test. A p-value greater than 0.01 indicates that the generated key sequence is random with 99% confidence. Table I lists the results of the tests for four different experimental scenarios. All results pass the tests, indicating that the generated secret keys are random with 99% confidence for practical key generation. Moreover, we evaluate the randomness of the final key sequence generated by both algorithms with channel reuse no more than three times (i.e., BM-SKGr and SA-SKGr). As shown in Table I, BM-SKGr and SA-SKGr pass all of the randomness tests.

D. Security Performance

Coping With Predictable Channel Attack: Channel measurements are predominantly influenced by LOS signals. If an attacker moves along a planned trajectory to obstruct the LOS between Alice and Bob, the channel measurements will become predictable when both Alice and Bob are stationary, thereby making the generated keys easy to predict. In our study, we

TABLE I
NIST STATISTICAL TEST RESULTS

BM-SKG / BM-SKGr / SA-SKG / SA-SKGr	Static Indoor	Mobile Indoor	Static Outdoor	Mobile Outdoor
Monobit Frequency	0.79 / 0.31 / 0.05 / 0.03	0.15 / 0.19 / 0.04 / 0.10	0.05 / 0.02 / 0.09 / 0.06	0.57 / 0.02 / 0.08 / 0.03
Block Frequency	0.40 / 0.37 / 0.88 / 0.15	0.24 / 0.24 / 0.92 / 0.99	0.29 / 0.91 / 0.74 / 0.33	0.62 / 0.62 / 0.62 / 0.44
Cumulative Sums (Fwd)	0.44 / 0.05 / 0.10 / 0.22	0.51 / 0.02 / 0.37 / 0.06	0.07 / 0.06 / 0.66 / 0.37	0.24 / 0.03 / 0.13 / 0.04
Cumulative Sums (Rev)	0.74 / 0.04 / 0.05 / 0.09	0.10 / 0.02 / 0.87 / 0.18	0.11 / 0.02 / 0.14 / 0.33	0.04 / 0.02 / 0.88 / 0.14
Runs	0.64 / 0.04 / 0.13 / 0.30	0.25 / 0.09 / 0.30 / 0.22	0.60 / 0.29 / 0.42 / 0.47	0.18 / 0.16 / 0.76 / 0.19
Longest Run of Ones	0.57 / 0.46 / 0.37 / 0.85	0.55 / 0.42 / 0.91 / 0.83	0.98 / 0.17 / 0.78 / 0.68	0.11 / 0.76 / 0.92 / 0.99
FFT	0.52 / 0.83 / 0.05 / 0.04	0.47 / 0.33 / 0.13 / 0.72	0.21 / 0.42 / 0.24 / 0.10	0.03 / 0.06 / 0.67 / 0.57
Approximate Entropy	0.74 / 0.72 / 0.35 / 0.17	0.36 / 0.40 / 0.57 / 0.66	0.28 / 0.13 / 0.91 / 0.60	0.58 / 0.88 / 0.64 / 0.38
Serial	0.64 / 0.62 / 0.33 / 0.72	0.66 / 0.25 / 0.91 / 0.42	0.49 / 0.66 / 0.49 / 0.02	0.32 / 0.42 / 0.40 / 0.09

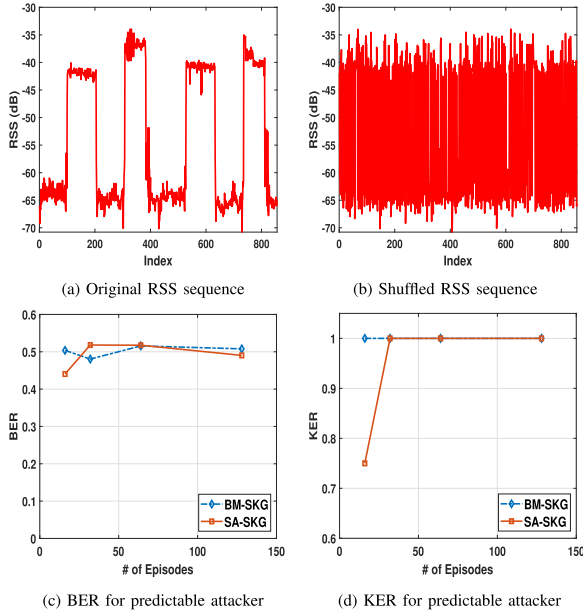


Fig. 14. Coping with predictable channel attack.

test our algorithms under such an attack by performing experiments with two stationary laptops acting as Alice and Bob. A volunteer is asked to periodically block the LOS channel of Alice and Bob for 60 sec. As shown in Fig. 14(a), the channel measurements observed by Alice have a clearly changing pattern corresponding to the LOS blockage. Fig. 14(b) shows that there are no observable patterns in the channel measurements after shuffling, indicating that our algorithms do not reveal the original channel measurement sequence. Even if the attacker can infer when specific RSS measurements are collected, it is still difficult to recover the channel measurement sequence with the correct shuffling order. As depicted in Fig. 14(c) and (d), the BERs of BM-SKG and SA-SKG are both near 50% for the attacker, while the attacker's KERs approach 100% when the number of episodes exceeds 16. This is because the random episode permutation or the random editing processes disrupt the changing pattern of the original channel measurements. Therefore, the quantitative analysis demonstrates that our BM-SKG and SA-SKG algorithms can defeat predictable channel attacks.

Coping With Stalking Attack: Intuitively, if an attacker follows Alice's or Bob's trajectory, the sequence of channel measurements collected by the attacker will be similar to those

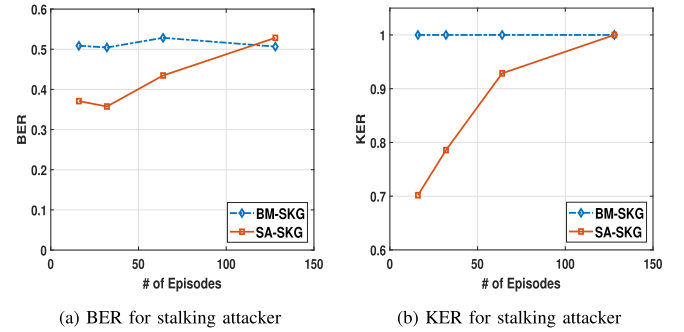


Fig. 15. Coping with stalking attack.

obtained by Alice or Bob. However, such an attacker cannot be very close to either Alice or Bob (at least half of the wavelength), as this would risk exposing the attacker. Consequently, the attacker observes a wireless channel independent of Alice and Bob's channel, even though the attacker follows the user's trajectory. Accordingly, the attacker cannot obtain the same sequence of channel measurements as Alice or Bob does. Moreover, random shuffling in our algorithms disrupts the order of the collected sequence, amplifying the difference between the sequence obtained by the attacker and the sequence received by the user. We ask a volunteer to follow Alice from a short distance and collect channel measurements from Bob when Alice and Bob are exchanging probe packets. We use the channel measurements collected by the attacker to generate secret keys and match them with the keys generated by Bob. As shown in Fig. 15, the stalker's BERs of BM-SKG remain above 50% as the number of episodes increases from 16 to 128. Similarly, the BERs of SA-SKG for the stalker increase to over 50% when the episode number is greater than 64. Notably, both the BM-SKG and SA-SKG algorithms have KERs of 100% when the episode number is around 128, showing that the stalker could no longer guess any complete key.

VII. CONCLUSION

In this work, we proposed two practical algorithms, BM-SKG and SA-SKG, for establishing a shared key with user-specified content between two users. Particularly, both algorithms enable one user to preset the secret key and transmit encrypted messages during channel sampling, which can reduce communication costs in key generation. BM-SKG achieved fast key generation

by designing an improved minimum-weight bipartite graph matching algorithm. SA-SKG ensured accurate and efficient key generation by calculating the edit distance and identifying the editing pattern. Instead of the error-prone quantization method commonly used in conventional approaches, BM-SKG and SA-SKG facilitated secret key agreement by matching the manipulated channel measurements between the two users. The generated secret key was derived based on permutation order and editing patterns with bit error rates as low as zero. Besides, we present an information reconciliation scheme and a channel reuse mechanism to ensure error-free key agreement with boosted key generation rates and high randomness. Our security analysis and performance evaluation based on extensive experiments in various scenarios demonstrated the effectiveness and efficiency of BM-SKG and SA-SKG. Our results showed that BM-SKG and SA-SKG achieve over 128-bit-equivalent cryptographic strength in around 27 msec and 9 msec, respectively.

REFERENCES

- [1] Y. Zou, J. Zhu, X. Wang, and L. Hanzo, "A survey on wireless security: Technical challenges, recent advances, and future trends," *Proc. IEEE*, vol. 104, no. 9, pp. 1727–1765, Sep. 2016, doi: [10.1109/JPROC.2016.2558521](#).
- [2] W. Stallings, *Cryptography and Network Security: Principles and Practice*, 6th ed. Englewood Cliffs, NJ, USA: Prentice Hall Press, 2013.
- [3] J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed. Boca Raton, FL, USA: CRC Press, Dec. 2020.
- [4] K. Zeng, "Physical layer key generation in wireless networks: Challenges and opportunities," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 33–39, Jun. 2015, doi: [10.1109/MCOM.2015.7120014](#).
- [5] S. Mathur, W. Trappe, S. Jana, C. Ye, and A. Reznik, "Radio-telepathy: Extracting a secret key from an unauthenticated wireless channel," in *Proc. Annu. Int. Conf. Mobile Comput. Netw.*, 2008, pp. 128–139, doi: [10.1145/1409944.1409960](#).
- [6] N. Patwari, J. Croft, S. Jana, and S. Kasera, "High-rate uncorrelated bit extraction for shared secret key generation from channel measurements," *IEEE Trans. Mobile Comput.*, vol. 9, no. 1, pp. 17–30, Jan. 2010, doi: [10.1109/TMC.2009.88](#).
- [7] K. Zeng, D. Wu, A. Chan, and P. Mohapatra, "Exploiting multiple-antenna diversity for shared secret key generation in wireless networks," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2010, pp. 1–9, doi: [10.1109/INFCOM.2010.5462004](#).
- [8] H. Liu, Y. Wang, J. Yang, and Y. Chen, "Fast and practical secret key extraction by exploiting channel response," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2013, pp. 3048–3056, doi: [10.1109/INFCOM.2013.6567117](#).
- [9] W. Xi et al., "Instant and robust authentication and key agreement among mobile devices," in *Proc. ACM Conf. Comput. Commun. Secur.*, 2016, pp. 616–627, doi: [10.1145/2976749.2978298](#).
- [10] X. Zhu, F. Xu, E. Novak, C. C. Tan, Q. Li, and G. Chen, "Using wireless link dynamics to extract a secret key in vehicular scenarios," *IEEE Trans. Mobile Comput.*, vol. 16, no. 7, pp. 2065–2078, Jul. 2017, doi: [10.1109/TMC.2016.2557784](#).
- [11] Z. Li, Q. Pei, I. Markwood, Y. Liu, and H. Zhu, "Secret key establishment via RSS trajectory matching between wearable devices," *IEEE Trans. Inf. Forensics Secur.*, vol. 13, no. 3, pp. 802–817, Mar. 2018, doi: [10.1109/TIFS.2017.2768020](#).
- [12] G. Margelis, X. Fafoutis, G. Oikonomou, R. Piechocki, T. Tryfonas, and P. Thomas, "Efficient DCT-based secret key generation for the Internet of Things," *Ad Hoc Netw.*, vol. 92, Sep. 2019, Art. no. 101744, doi: [10.1016/j.adhoc.2018.08.014](#).
- [13] G. Li, A. Hu, J. Zhang, L. Peng, C. Sun, and D. Cao, "High-agreement uncorrelated secret key generation based on principal component analysis preprocessing," *IEEE Trans. Commun.*, vol. 66, no. 7, pp. 3022–3034, Jul. 2018, doi: [10.1109/TCOMM.2018.2814607](#).
- [14] M. Bloch, J. Barros, M. Rodrigues, and S. McLaughlin, "Wireless information-theoretic security," *IEEE Trans. Inf. Theory*, vol. 54, no. 6, pp. 2515–2534, Jun. 2008, doi: [10.1109/TIT.2008.921908](#).
- [15] D. Chen, Z. Qin, X. Mao, P. Yang, Z. Qin, and R. Wang, "Smokegrenade: An efficient key generation protocol with artificial interference," *IEEE Trans. Inf. Forensics Secur.*, vol. 8, no. 11, pp. 1731–1745, Nov. 2013, doi: [10.1109/TIFS.2013.2278834](#).
- [16] D. Guo, K. Cao, J. Xiong, D. Ma, and H. Zhao, "A lightweight key generation scheme for the Internet of Things," *IEEE Internet Things J.*, vol. 8, no. 15, pp. 12 137–12 149, Aug. 2021, doi: [10.1109/JIOT.2021.3060438](#).
- [17] H. Liu, Y. Wang, Y. Ren, and Y. Chen, "Bipartite graph matching based secret key generation," in *Proc. IEEE Conf. Comput. Commun.*, 2021, pp. 1–10, doi: [10.1109/INFOCOM42981.2021.9488848](#).
- [18] S. Jana, S. N. Premnath, M. Clark, S. K. Kasera, N. Patwari, and S. V. Krishnamurthy, "On the effectiveness of secret key extraction from wireless signal strength in real environments," in *Proc. Annu. Int. Conf. Mobile Comput. Netw.*, 2009, pp. 321–332, doi: [10.1145/1614320.1614356](#).
- [19] H. Zhao, Y. Zhang, X. Huang, Y. Xiang, and C. Su, "A physical-layer key generation approach based on received signal strength in smart homes," *IEEE Internet Things J.*, vol. 9, no. 7, pp. 4917–4927, Apr. 2022, doi: [10.1109/JIOT.2021.3119053](#).
- [20] M. S. Kumar, R. Ramanathan, M. Jayakumar, and D. K. Yadav, "Physical layer secret key generation using discrete wavelet packet transform," *Ad Hoc Netw.*, vol. 118, Jul. 2021, Art. no. 102523, doi: [10.1016/j.adhoc.2021.102523](#).
- [21] S. Fang, I. Markwood, and Y. Liu, "Wireless-assisted key establishment leveraging channel manipulation," *IEEE Trans. Mobile Comput.*, vol. 20, no. 1, pp. 263–275, Jan. 2021, doi: [10.1109/TMC.2019.2939529](#).
- [22] N. Aldaghri and H. Mahdavi, "Fast secret key generation in static environments using induced randomness," in *Proc. IEEE Glob. Commun. Conf.*, 2018, pp. 1–6, doi: [10.1109/GLOCOM.2018.8647945](#).
- [23] N. Aldaghri and H. Mahdavi, "Physical layer secret key generation in static environments," *IEEE Trans. Inf. Forensics Secur.*, vol. 15, pp. 2692–2705, 2020, doi: [10.1109/TIFS.2020.2974621](#).
- [24] G. Li, H. Yang, J. Zhang, H. Liu, and A. Hu, "Fast and secure key generation with channel obfuscation in slowly varying environments," in *Proc. IEEE Conf. Comput. Commun.*, 2022, pp. 1–10, doi: [10.1109/INFCOM48880.2022.9796694](#).
- [25] Q. Wang, H. Su, K. Ren, and K. Kim, "Fast and scalable secret key generation exploiting channel phase randomness in wireless networks," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2011, pp. 1422–1430, doi: [10.1109/INFCOM.2011.5934929](#).
- [26] R. Wilson, D. Tse, and R. Scholtz, "Channel identification: Secret sharing using reciprocity in ultrawideband channels," *IEEE Trans. Inf. Forensics Secur.*, vol. 2, no. 3, pp. 364–375, Sep. 2007, doi: [10.1109/TIFS.2007.902666](#).
- [27] B. Azimi-Sadjadi, A. Kiai, A. Mercado, and B. Yener, "Robust key generation from signal envelopes in wireless networks," in *Proc. ACM Comput. Commun. Secur. Conf.*, 2007, pp. 401–410, doi: [10.1145/1315245.1315295](#).
- [28] Y. Liu, S. C. Draper, and A. M. Sayeed, "Exploiting channel diversity in secret key generation from multipath fading randomness," *IEEE Trans. Inf. Forensics Secur.*, vol. 7, no. 5, pp. 1484–1497, Oct. 2012, doi: [10.1109/TIFS.2012.2206385](#).
- [29] W. Xi, M. Duan, X. Bai, K. Zhao, L. Mo, and J. Zhao, "KEEP: Secure and efficient communication for distributed IoT devices," *IEEE Internet Things J.*, vol. 8, no. 16, pp. 12758–12770, Aug. 2021, doi: [10.1109/JIOT.2020.3011558](#).
- [30] J. Zhang, M. Ding, D. López-Pérez, A. Marshall, and L. Hanzo, "Design of an efficient OFDMA-based multi-user key generation protocol," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8842–8852, Sep. 2019, doi: [10.1109/TVT.2019.2929362](#).
- [31] C. Hansen, Q. Wei, J.-S. Shieh, P. Fourcade, B. Isableu, and L. Majed, "Sample entropy, univariate, and multivariate multi-scale entropy in comparison with classical postural sway parameters in young healthy adults," *Front. Hum. Neurosci.*, vol. 11, Apr. 2017, Art. no. 206, doi: [10.3389/fnhum.2017.00206](#).
- [32] M. L. Fredman and R. E. Tarjan, "Fibonacci heaps and their uses in improved network optimization algorithms," *J. ACM*, vol. 34, no. 3, pp. 596–615, 1987, doi: [10.1145/28869.28874](#).
- [33] P. Erdős and A. Rényi, "On random matrices II," *Studia Scientiarum Mathematicarum Hungarica*, vol. 3, pp. 459–464, 1968.
- [34] E. Ukkonen, "On approximate string matching," in *Proc. Int. Conf. Fundam. Comput. Theory*, Springer, 1983, pp. 487–495, doi: [10.1007/3-540-12689-9_129](#).
- [35] U. Maurer and S. Wolf, "Secret-key agreement over unauthenticated public channels-part II: Privacy amplification," *IEEE Trans. Inf. Theory*, vol. 49, no. 4, pp. 839–851, Apr. 2003, doi: [10.1109/TIT.2003.809559](#).

- [36] B. G. Staff, L. Books, and B. Group, *Permutations: Shuffling, Permutation, Parity of a Permutation, Transposition Cipher, Cayley's Theorem, Stirling Number, Landau's Function*. Memphis, TN, USA: General Books, 2010.
- [37] H. Liu, J. Yang, Y. Wang, and Y. Chen, "Collaborative secret key extraction leveraging received signal strength in mobile wireless networks," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2012, pp. 927–935, doi: [10.1109/IN-FCOM.2012.6195843](https://doi.org/10.1109/IN-FCOM.2012.6195843).
- [38] Texas Instruments CC13x0, CC26x0 SimpleLink Wireless MCU. Accessed: 2020. [Online]. Available: <https://www.tij.co.jp/lit/ug/swcu117i/swcu117i.pdf>
- [39] A. Rukhin, J. Soto, J. Nechvatal, M. Smid, and E. Barker, "A statistical test suite for random and pseudorandom number generators for cryptographic applications," Nat. Inst. Standard Technol., Gaithersburg, MD, USA, Tech. Rep. 800-22, 2010.

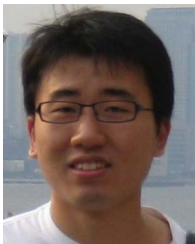


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